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P-XX

Energy-Aware Arm and Base Selection for a Dual-Gantry Quad-Arm Ceiling Robot Using a GNG Capability Map

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Which arm should act, and where should its base go?
One energy cost J, one base-aware capability map, both answers at once.

Social Background

An aging population is driving demand for assistive robots that work **inside the home**: fetching, tidying, delivering.

Homes are small, cluttered, and **shared with residents**, so a floor-driving mobile manipulator gets in the way.

Putting the robot on the ceiling keeps the floor free: it assists from above without blocking the occupant.

Platform. Two ceiling rails (gantries), **two Kinova Gen3 Lite arms each** $\rightarrow$ four arms.

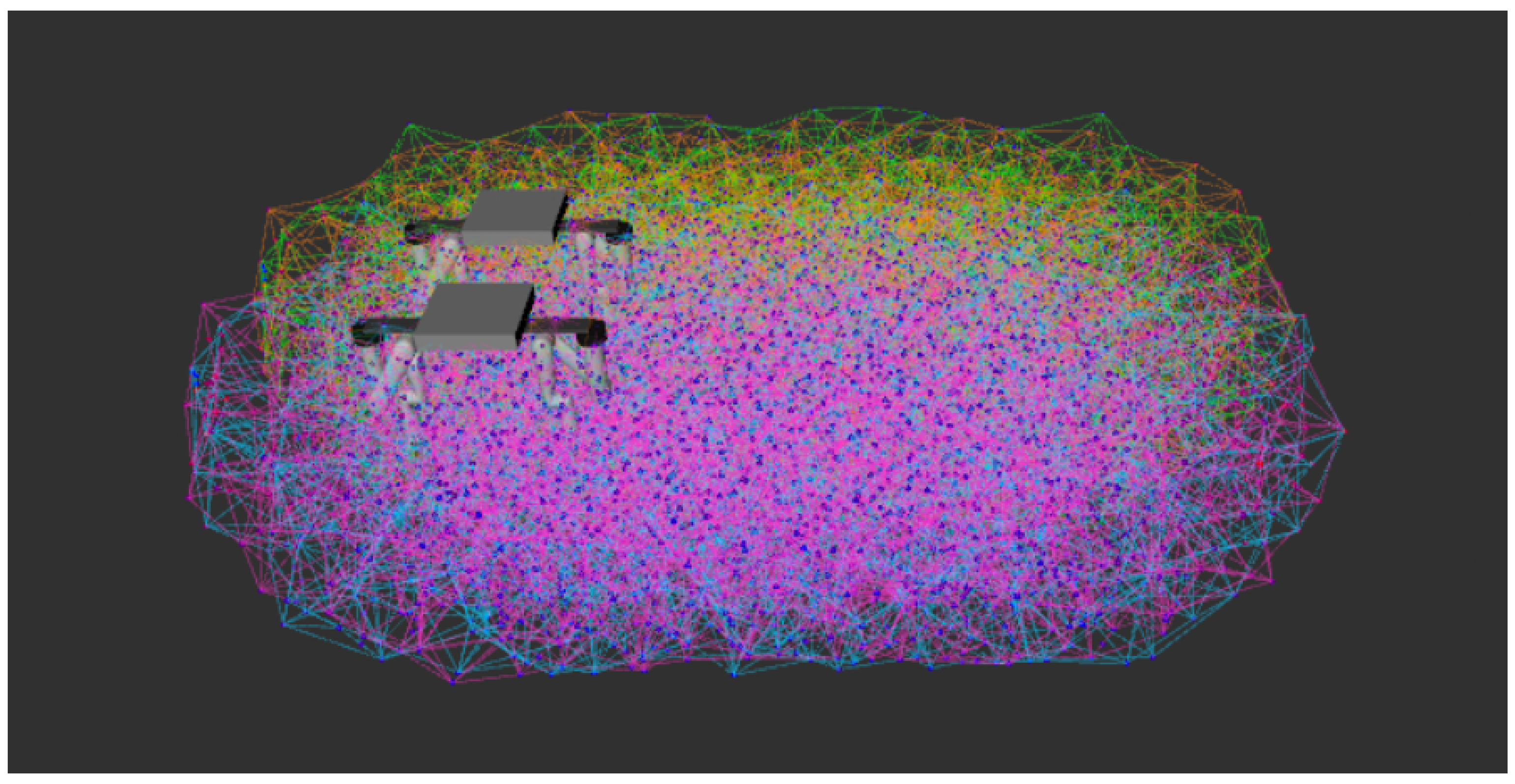
Each rail adds a prismatic axis d and a revolute axis θ that its two arms *share*, so every arm is an **8-DOF** chain $\mathbf{q} = [d, \theta, q_1 \dots q_6]$ and moving the base moves two arms at once.

Problems

- **P1 — Two coupled decisions.** Each pick asks *which arm* acts *and where its base stops.* Moving the heavy base is a dominant energy cost.
- **P2 — Maps assume a fixed base.** Classic reachability/capability maps are built for one arm on a stationary base, so they never see the workspace that base motion creates.
- **P3 — A cell is only one bit.** Maps store *whether* a cell is reachable, not at what cost or from which configuration: no IK seed, no arm choice.

Core Concept: Base-Aware GNG Capability Map

- Sample the **full 8-DOF** $\mathbf{q} = [d, \theta, q_1 \dots q_6]$ inside joint limits, take forward kinematics (Pinocchio), record manipulability $w = \sqrt{\det(JJ^\top)}$.
- Fit a **Growing Neural Gas** net: each node stores $[\mathbf{x} \mid \mathbf{q}]$ — a reachable point *and* a representative configuration reaching it, **rail placement included**.
- Winner search uses the *task* part only (nodes tile the workspace); adaptation updates the whole vector, so a node's $\mathbf{q}$ stays one valid posture, not an average of incompatible base positions.
- **Boundary seeding**: a shell of nodes is pinned on the measured reach surface *before* the interior grows, keeping the map faithful at the workspace edge.
- One map per arm $\rightarrow$ four maps, queried independently at selection time.



Base-aware GNG map of one arm (RViz): nodes coloured by manipulability plus the pinned boundary shell on the true rail-extended reach surface. The elongated extent is the rail sweeping x.

E0 — map built per arm: 80,000 FK samples $\rightarrow$ **2915 nodes** (600 pinned on the boundary shell), 14,547 edges, median node spacing **0.090 m**, and a full **8-DOF** $\mathbf{q}$ stored on every node.

Energy-Aware Selection Cost J

$$J = w_{gl}\frac{\Delta_{lin}}{\rho_{gl}} + w_{gr}\frac{\Delta_{rot}}{\rho_{gr}} + w_{arm}\frac{\Delta_{arm}}{\rho_{arm}} + w_{dist}\frac{e}{\rho_{dist}} - w_{manip}\frac{m}{\rho_{manip}}$$

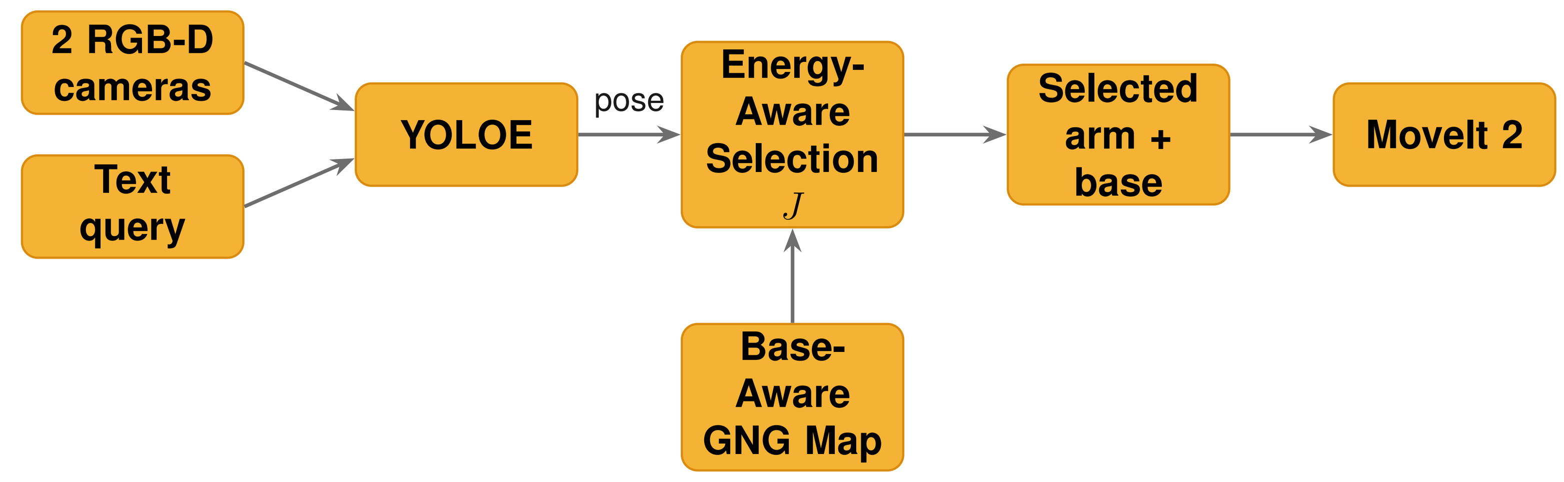
- $\Delta_{lin}, \Delta_{rot}, \Delta_{arm}$: rail-linear, rail-rotation and summed arm-joint travel from the current state; e: tool-to-object distance; m: manipulability, entering *negatively* — dexterous is cheap.
- Each term is divided by a reference ρ (its median over a calibration set), so a weight reads as a priority, not a unit artefact.
- Weights are **fit by OLS** against the *measured* mechanical energy of 168 executed picks, not hand-tuned.

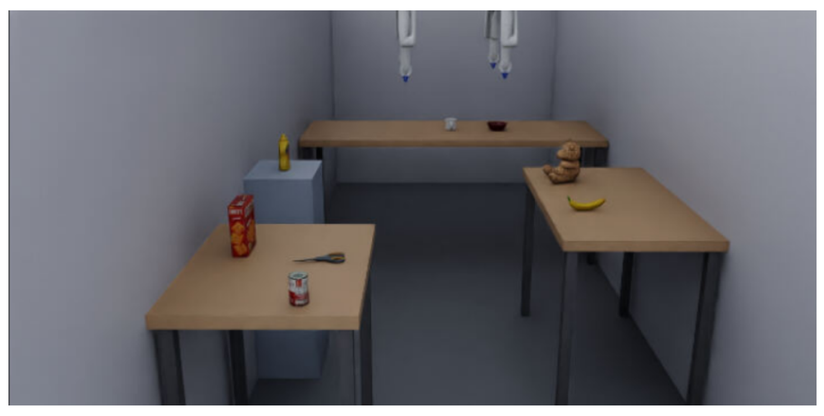
Term	Symbol	Weight	Ref. ρ
Rail linear travel	Δ_{lin}	**27.16**	1.18
Tool-to-object distance	e	12.78	1.45
Rail rotation travel	Δ_{rot}	3.08	1.24
Manipulability (subtr.)	m	2.95	0.105
Arm joint travel	Δ_{arm}	0.09	7.18

Calibrated weights: $R^2 = 0.857$, Spearman$(J, E) = 0.909$. Rail travel dominates once rail friction is modelled.

Selection. Pool map nodes within a spacing-scaled radius of the target *from all four arms*, score with J, sort, and try them **round-robin across arms**. Each candidate seeds IK from its own $\mathbf{q}$; MoveIt 2 plans; the first collision-free plan wins. The search is **plan-only** — the arm never moves while deciding.

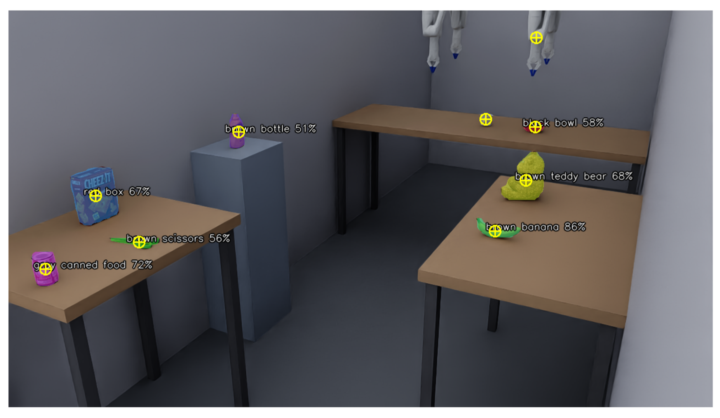
System and Pipeline





All experiments run in **NVIDIA Isaac Sim**, a digital twin of our trailer living laboratory built from the *same URDF* as the real robot. Two overhead RGB-D cameras are the only sensor: colour feeds detection, depth builds the collision octomap.

Perception Front End

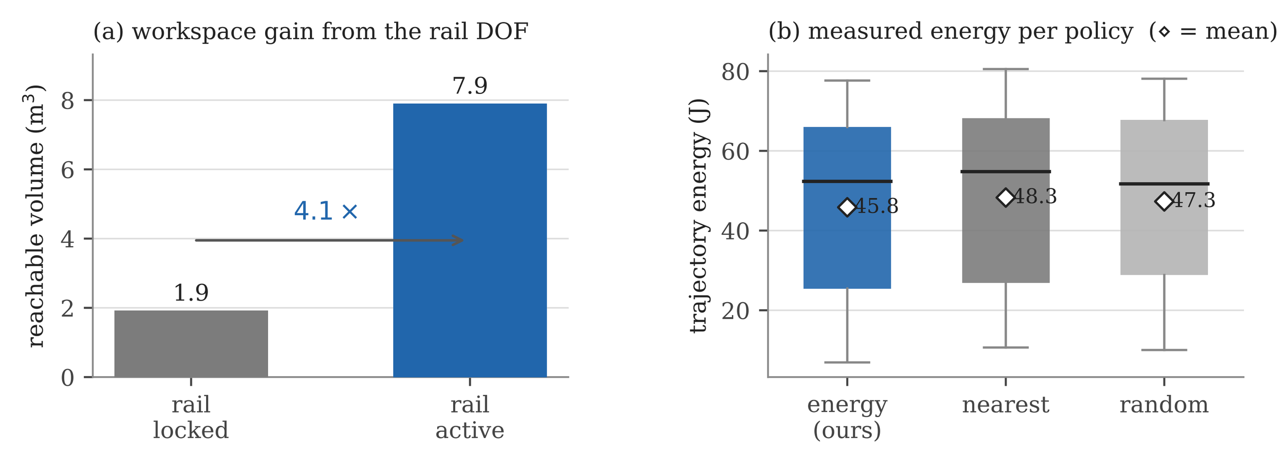


YOLOE segments each object from a free-form text prompt. Masked depth is back-projected and the *median* world point taken as the centroid; the two cameras are fused by centroid proximity (0.10 m) and majority label voting. No fixed object database: ask by name.

E4 — YOLOE vs. Isaac ground truth (8 objects, 483 frames): **100%** detection (3864/3864), mean localization error **0.023 m**, **0%** false positives at conf $= 0.25$.

Experiments and Results

E1 — the rail is worth $4\times$ the workspace. Rail locked (6 arm DOF) vs. rail sampled over its full travel, at matched sample *density*: reachable volume grows **1.920→7.899 m^3** at 0.05 m voxels ($\approx 4\times$). **E1b — boundary seeding** cuts the node-hull shortfall against the true FK reach surface from **0.221 m to 0.008 m** (along the rail axis 0.501 m $\rightarrow$ 0.017 m).



(a) Reachable volume, rail locked vs. active. (b) Measured trajectory energy per multi-arm policy (◇ = mean).

E3 — selection policies, 60 target poses, plan-only, mean/median over successful picks, under a rail friction model with assumed Coulomb/viscous terms (a modelling assumption, not measured hardware data).

Mode	Success	Rail lin. (m)	Energy (J)
energy (ours)	**56/60 (93.3%)**	**0.97**/1.01	**45.84**/52.34
nearest	56/60 (93.3%)	1.19/1.41	48.28/54.79
random	56/60 (93.3%)	1.17/1.28	47.31/51.72
fixed (best, arm_4)	39/60 (65.0%)	1.14/1.21	49.79/55.78
fixed (worst, arm_1)	38/60 (63.3%)	1.13/1.28	45.43/43.12

Fixed-arm energies are computed over that arm's own smaller reachable set, so they are not comparable at equal coverage.

- **Coverage:** every multi-arm policy reaches **93.3%** plan success vs. **63–65%** for any single fixed arm.
- **Energy:** ours has the **lowest mean measured energy** of the multi-arm policies (**45.84 J** vs. 48.28 J nearest, -5.1%; 47.31 J random, -3.1%), winning **57–61%** of same-target pairwise comparisons — a real but *partial* win: medians are closer and the distributions overlap.
- **The intended trade appears:** ours spends *more* arm travel to avoid costlier rail travel — one logged pick took arm_3 (0.64 m rail, 42.4 J) over the nearest arm_4 (1.15 m rail, 72.5 J): **30.1 J** saved.

End-to-End Validation

E6: perception $\rightarrow$ localization $\rightarrow$ energy-aware selection $\rightarrow$ IK $\rightarrow$ planning $\rightarrow$ real Isaac physics execution, on the 7 reachable objects $\times$ 5 trials: **35/35 (100%)** picks, no failure in any category, mean time-to-pick **39.6 s**.

E5: the map's nearest-node test classifies reachable vs. unreachable objects **100% (8/8)** at `reach_radius` ≥ 0.12 m, zero false positives.

Summary and Contributions

- An **energy-aware cost** J that picks the arm *and* the full 8-DOF configuration, rail travel included, instead of the nearest reachable pose.
- A **base-aware GNG capability map** treating the ceiling rail as a genuine DOF: every node carries its own rail placement and manipulability, accurate to **centimetres** at the workspace edge.
- An **Isaac Sim evaluation**: $\approx 4\times$ workspace gain, 93.3% vs. 63–65% plan success, lowest mean measured energy among multi-arm policies, 35/35 end-to-end picks.

Future work. All results are in simulation. Next: validation on the physical ceiling-rail platform, and how the sim-to-real gap shifts the calibrated weights of J.